

CHAPTER 4

ACS system design

4.1. Introduction

This chapter describes the process of creating a preliminary control-system design. It describes how to interpret customer requirements, how to select actuators and sensors, and how to organize your design. Designing an attitude control system from scratch presents the control-systems engineer with a bewildering set of choices. The engineer must select:

- Spacecraft type (dual-spin, momentum-bias, zero-momentum, Sun-nadir, etc.);
- Control strategy (active or passive);
- Actuators (wheels, magnetic torquers, thrusters, etc.);
- Sensors (Earth, star, Sun, planet, magnetic field, gyros etc.);
- Processors (1750A, 80386, 1802, 8086, BAE RAD750, Microsemi SAMRH71 etc.);
- Delta-V engine for orbit changes (Apogee Kick Motor (AKM), Liquid Apogee Engine (LAE), etc.);
- Station-keeping thrusters (Rocket Engine Assembly (REA), Electrothermal Hydrazine Thruster (EHT)s, Arc jets, Ion, etc.);
- Interfaces (Analog-to-Digital (A/D) and Digital-to-Analog (D/A) converters, multiplexers, data buses, etc.).

You will note that many terms end with “assembly.” This is because the device is a combination of electronics, optics, mechanisms, etc. that together form an assembly. This may just be RCA Astroelectronics nomenclature, but it is used throughout this book. Within each of these categories there are many choices. For example, star sensors can be trackers, cameras, or mappers and may have built-in star identification or may pass a pixel map to the spacecraft computer. Sun sensors can be digital or analog, have one or two axis of sensing and may be high or low accuracy. All of these items must be chosen in conjunction with the other subsystem engineers and the program-management office. Often the payload dictates the spacecraft type.

This section gives a qualitative discussion of attitude control system design. Its purpose is to outline an approach to designing a spacecraft control system and provide the reader with an introduction to some of the options.

4.2. Design flow

Fig. 4.1 shows a typical spacecraft-design flow with some options.

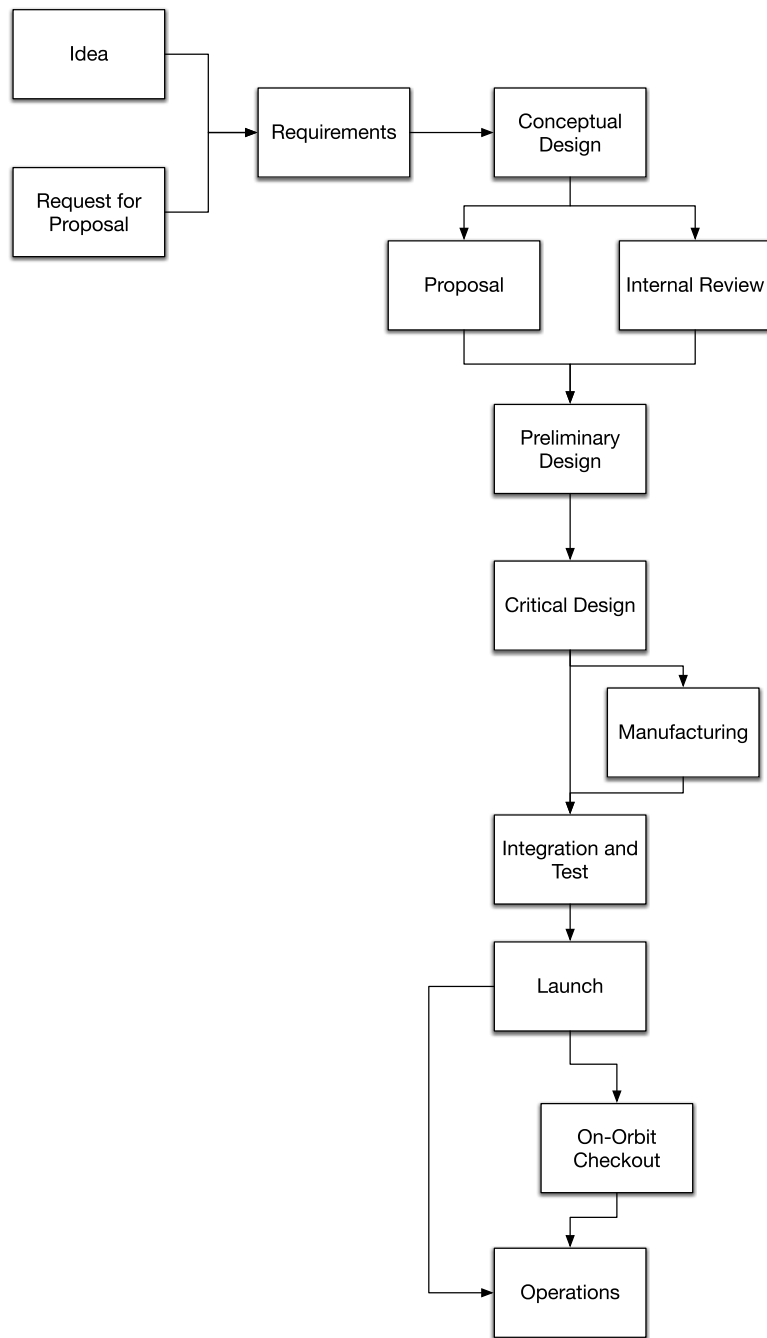


Figure 4.1 Spacecraft-design flow.

Table 4.1 gives system engineering and proposal terms. Not every spacecraft goes through these stages or has all of these reviews. Some may have more.

Table 4.1 System-engineering terms.

Term	Definition
CDR	Critical Design Review
LRR/FRR	Launch/Flight Readiness Review
MRR	Manufacturing Readiness Review
PDR	Preliminary Design Review
RFI	Request for Information that may lead to an RFP
RFP	Request For Proposal
PTR	Pink Team Review
RTR	Red Team Review

4.3. Organization of ACS design teams

An attitude control system consists of hardware, algorithms, and software. The algorithms are typically embedded in the software. They take data from sensors and process it to produce the commands to the actuators to accomplish the control objectives. Different engineers take on different roles in this process. Fig. 4.2 shows one decomposition of the roles for an ACS design. We distinguish between the ACS Controls Engineer and Systems Engineer. The former is responsible for doing all of the analysis and design of the ACS system. The latter does requirements analyses, interacts with the hardware vendors and program offices and also with the other subsystem engineers. Simulation may be a produce of the ACS analysts or be done by the software group. The software engineers implement the ACS algorithms in the flight software and write the interface software that connects the algorithms to the hardware. ACS engineers interact with propulsion if thrusters are used for attitude control. This diagram shows only the major roles and connections.

4.4. Requirements analysis

4.4.1 Direct requirements

Each mission will have a set of requirements. Requirements are either direct or derived. Typical direct mission requirements are for:

- Pointing accuracy;
- Pointing knowledge;
- Jitter;
- Lifetime;

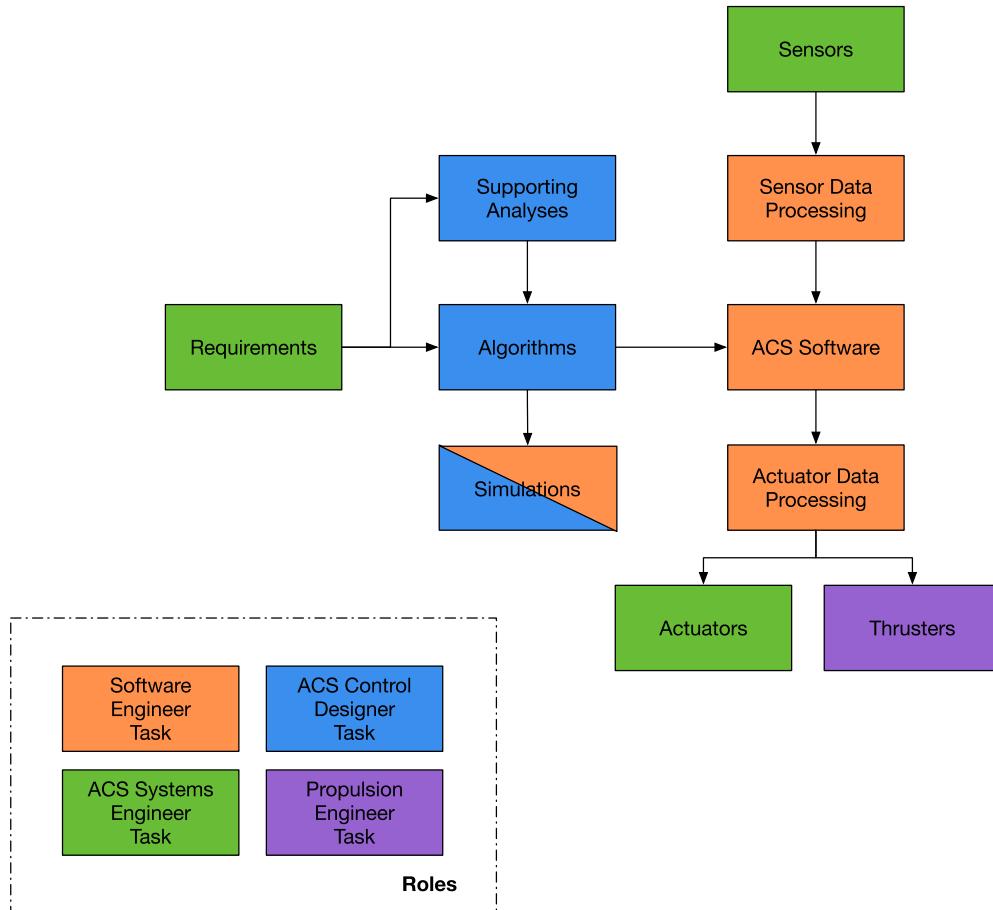


Figure 4.2 Spacecraft control-system design flow.

- Reliability;
- Redundancy.

Pointing accuracy is how tightly the spacecraft points at its target and knowledge is how well you know the attitude error. For example, a passively stabilized spacecraft with a star sensor might have very high attitude knowledge but relatively poor pointing accuracy. Jitter is measured in a variety of ways but ultimately it is a measure of how much the attitude changes in a given period of time that is usually not controllable by the pointing-control system. For example, if you use noisy sensors and low-resolution D/A (digital-to-analog) converters, you are likely to have a lot of jitter. Jitter is attitude motion that is not controllable by the control system. It may be beyond the bandwidth of the controller or hidden by a notch filter in the control loop.

The pointing numbers may be given for the spacecraft as a whole or for individual payloads. If the latter is the case, you have the additional option of putting some of the payloads on articulated platforms with their own sensors to improve the pointing of those payloads. For example, on a dual-spin spacecraft only the despun platform has a pointing requirement. The spinning part is only required to store momentum (and sometimes generate power). Pointing requirements only apply to the payload. It is not unusual to collocate the payload and attitude control sensors on the same composite structure (i.e., structurally stiff with low coefficients of thermal expansion) to minimize errors due to structural vibration and thermal distortion.

The lifetime requirement stipulates that the spacecraft must have a certain probability of lasting for the desired life-span. Two things can happen to terminate a mission. The spacecraft can run out of consumables (usually fuel but sometimes coolant for payloads), or the components can fail or wear out.

Redundancy requirements are often given as how long it takes the spacecraft to return to normal operation after a failure. The assumption is that no one failure will impact the mission, i.e., cause it not to meet its requirements, except for a permitted transient period after the failure. Different failures may have different requirements. Commercial customers usually require the quickest recovery from a failure since they lose money if the satellite is out of operation, not to mention irritating millions of customers whose sports broadcast is suddenly interrupted. Scientific customers are usually more lenient. The Hubble Space Telescope was operating out of specification for a long time before it was fixed.

The customer does not always specify the requirements carefully and may change them without notice as the design evolves. Thus, aside from meeting the baseline requirements, it is always necessary to make the design robust enough to tolerate requirements changes.

Another problem with requirements is that the customer may specify requirements in a way that needlessly constrains the design. For example, a geosynchronous-satellite customer may specify the Euler angles for roll, pitch, and yaw (used to measure spacecraft orientation) when they really mean to specify beam pointing (circular error) or even better, power distributed over the coverage area. If you are given the third type of requirement you could meet it by increasing the beam power. If you are given the second you could trade roll accuracy against pitch accuracy and ignore yaw altogether (with the caveat mentioned above).

4.4.2 Indirect (or derived) requirements

Derived requirements are levied on the ACS subsystem by other subsystems. If the power over the coverage area is the direct requirement then attitude accuracy becomes an indirect requirement. Power consumption of your components is always an indirect requirement. Some indirect requirements are:

- Power;
- Thermal;
- Radiation hardness;
- Radio-frequency emission sensitivity;
- Induced noise (such as from an unbalanced momentum wheel);
- Computational loading (i.e., how many floating-point operations you can use per control cycle).

4.4.3 Control-system requirements

Traditional control-system requirements, such as gain and phase margins, are rarely specified. They must be derived from the pointing and jitter requirements. For example, most satellites have a special thruster control mode. When you turn on the delta-V thrusters, there will be an attitude transient. This transient must not exceed the pointing requirements for the spacecraft; hence this puts a limit on the peak overshoot that is permissible. It is usually desirable not to have to change control-system gains over the mission life; hence this places a requirement on the tolerance of the control system to properties changes. Sensor-scale factors may drift due to accumulated radiation damage, leading to a requirement on the tolerance to scale-factor variations.

While some, such as sensor- and actuator-scale factors and inertia variations, can be lumped into an aggregate gain variation, it is a good idea to perform a sensitivity analysis on the closed-loop control system for all identifiable parameters that are known to vary.

4.5. Satellite design

4.5.1 Selecting a satellite configuration

Satellites generally have two major phases in their life. The first phase is transfer orbit during which the satellite is placed in its operational orbit. The second phase is the mission orbit where the satellite performs its nominal mission. Deep-space spacecraft may have long transfers. They often have separate scientific requirements during transfer to their final target. For example, New Horizons had requirements for its Pluto flyby, but also has requirements for its flight into interstellar space. Low-Earth orbiting satellites are often placed directly into their mission orbits and do not have a transfer-orbit phase.

The first step is choosing the mission-orbit configuration, followed by choosing the transfer-orbit configuration. A few of the more popular satellite configurations are:

- Gravity-gradient stabilized;
- Spinner;
- Dual-spin;
- Bias momentum;
- Zero momentum;
- Sun-nadir.

Gravity-gradient stabilization is passive and at most requires a libration damper. Libration dampers are dampers designed to damp spacecraft rigid-body oscillations due to gravity-gradient disturbances. If there is sufficient internal damping (due to heat pipes, fuel slosh, or structural damping), even the damper may not be required. Gravity-gradient stabilization is suitable only for spacecraft with loose pointing requirements that do not have to change their orientation.

A spinner can be used if one axis of the spacecraft is required to remain inertially fixed and the payload is not affected by spinning, for example a satellite with a long wire antenna along the spin axis. If the satellite spins about its major axis a passive damping system will suffice. If it has to spin about its minor axis an active nutation control system (to damp the natural oscillation of a spinning spacecraft at the nutation frequency) is needed to keep it out of a flat spin (i.e., a steady spin about any other axis). The satellite will also need a control system to keep the spin axis in the same inertial direction if there are significant inertially fixed torques, for example due to a center-of-solar-pressure offset from the center-of-mass along the spin axis.

Dual-spin spacecraft have two rotating parts. Usually one is a platform that rotates at orbit rate (and consequently is nominally pointing at the Earth). The other rotates at high speed and gives the spacecraft gyroscopic stiffness. A dual-spin spacecraft has a large structure on the spacecraft rotating. Often the solar cells are attached to the rotating part.

A bias-momentum spacecraft is conceptually the same as a dual-spin spacecraft, except that the part that is rotating at high speed is a momentum wheel and it rotates at very high rates, on the order of 6000 rpm. Its momentum may be large but the wheel's spin-axis moment of inertia is small. Both dual-spin and momentum-bias spacecraft have been used historically to keep costs down for commercial communication satellites since the gyroscopic stiffness helps to keep the spacecraft oriented. The coupling between the roll and yaw axes allows the elimination of the yaw sensor, which was the most expensive on the spacecraft since it requires sensing of the Sun.

A zero-momentum design has, ideally, zero net inertial momentum. The spacecraft may be controlled by reaction wheels, control-moment gyros, magnetic torquers, or by thrusters. With a spacecraft with momentum-exchange devices, they may each have significant momentum but the vector sum of their momentum is zero. It is relatively easy to rotate a spacecraft with zero momentum because there is no gyroscopic stiffness to resist rotation. However, if the control system fails, there is also no gyroscopic stiffness to keep the spacecraft from rotating and it may tumble. Magnetic torquers can only control two axes at a time but over an orbit can provide "average" three-axis control.

A Sun-nadir pointing spacecraft is a spacecraft that points the bus at nadir and the solar-array normal at the Sun. This type of spacecraft has only one degree of rotational freedom for the solar array. Thus to keep the arrays on the Sun the spacecraft must rotate about the axis pointing towards the Earth (yaw). This allows the spacecraft to produce more power than would be otherwise possible.

4.5.2 Selecting a control strategy

Passive control can be used on simple satellites that do not have tight pointing requirements. One strategy would be a gravity-gradient-stabilized satellite with a ball-in-tube libration damper. The damper damps out libration frequencies (the frequency of the gravity-gradient modes of oscillation). Active control requires either analog or digital electronics, sensors, and actuators and can add greatly to the cost of a satellite.

4.5.3 Selecting actuators

Typical satellite actuators are:

- Thrusters;
- Gimballed thrusters;
- Wheels or spheres (momentum or reaction);
- Pivoted wheels (control-moment gyros, for example);
- Magnetic torquers;
- Reflective surfaces (either to solar or aerodynamic drag).

We will discuss thrusters, wheels, and pivoted wheels further in this section. Additional details may be found in [Chapter 10](#).

4.5.4 Thrusters

If the satellite has a requirement to maintain its orbit it will require thrusters and a complete propulsion system. It is almost always necessary to use some thrusters for attitude control during orbit-adjustment maneuvers. This may be done by firing other thrusters to provide control torques to compensate for the disturbance torques generated by the delta-V thrusters, or by gimbaling the delta-V thrusters. For example, the Space Shuttle Orbiter does both. It gimbals the Orbital Maneuvering System (OMS) engines and also fires bipropellant thrusters. Gimbaling is generally not very popular because the gimbal mechanisms tend to be heavy, complex, and expensive. A gimbal was used on NASA's Deep Space 1 to point the ion engine through the center-of-mass, effectively providing 2-axis control.

Thrusters can be throttled, fired with a fixed-length pulse, or pulsewidth modulated. Pulsewidth modulation means that the thruster is on at a fixed thrust level for only part of the control period so that the average torque equals the commanded torque. This technique can be extended to have a variable firing period (known as pulsewidth–pulse-frequency modulation). Throttled thrusters, which can produce a variable magnitude of thrust, tend to be expensive and complex and are rarely used. Pulsewidth modulation is frequently used because it lends itself to linear control systems. Fixed pulsewidth–control systems are derived from optimal control theory and minimize fuel consumption. They are used on the Shuttle and were used on the Apollo spacecraft. However, they do not lend themselves to spacecraft that cannot be modeled as rigid bodies.

4.5.5 Wheels

A momentum wheel nominally has a fixed (large) momentum. A reaction wheel nominally has zero momentum. A reaction wheel (or reaction sphere) works by providing a reaction to the torque that the motor produces. The torque is applied equally to the spacecraft and to the wheel. Wheels present an interesting control problem. As they spin up they create a momentum bias on the spacecraft, thus changing the system dynamics. This perturbation is nonlinear because the torque on the spacecraft is the product of the spacecraft angular velocity and the wheel momentum. If the disturbances are small, and mostly cyclic, and the spacecraft has a relatively large inertia, the plant perturbation is second order. Many algorithms have been written to take advantage of this during maneuvers. The algorithm will try to pick a path in velocity space that minimizes time while staying within the momentum limits of the wheels.

4.5.6 Pivoted wheels and control-moment gyros

Pivoted wheels use pivots (either one or two) to generate transverse axis torques. They also use the wheel like a reaction wheel. Control-moment gyros (CMGs) use only the gimbal motors to provide torque and the wheel is maintained at a constant speed. CMGs come in double- and single-gimbaled types. The former require complex slip rings to get power to the inner gimbal. Since one CMG only provides at most two axes of torque, at least two are required. Generally, more are needed and it is necessary to distribute the control activity among the gimbals in an optimal way (usually to keep the CMG platforms at small angles relative to the spacecraft). CMGs are popular for maneuvering spacecraft of intermediate size or for really large spacecraft (like Skylab, the first American space station, the International Space Station, or certain Earth-observing systems).

A single pivoted wheel is shown in Fig. 4.3. Pivoted wheels are used on Lockheed Martin momentum-bias communications satellites and on the Hughes HS601 spacecraft. The former uses a single pivot about roll while the latter employs a double pivot.

Pivoted wheels and CMGs require momentum unloading unless the complementary filter approach is used. In either case, an actuator that can apply an external torque to the spacecraft, such as a thruster or magnetic torquer, is required. Reaction wheels, pivoted wheels, and CMGs can redistribute momentum within a spacecraft; they cannot change the total inertial momentum of the spacecraft or change its vector in the inertial frame. Many spacecraft with station-keeping requirements only unload momentum during stationkeeping maneuvers when it can be done “for free” using the station-keeping thrusters.

4.5.7 Selecting sensors

Typical satellite sensors are:

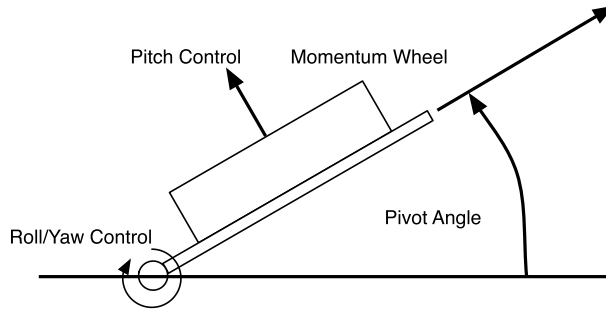


Figure 4.3 Pivoted wheel.

- Earth sensors (scanning or static);
- Star sensors (mappers or trackers);
- Sun sensors (one or two axis);
- Horizon sensors (need a spinning spacecraft);
- Gyros (many types);
- Magnetometers;
- Potentiometers;
- Angle encoders.

Many satellites are Earth or planet pointing and need to point a payload at a target on the Earth or a planet. Some are inertially pointing (such as the space telescope) and point to targets away from the Earth. Star sensors and Sun sensors give an inertial reference directly and can, through means of transformation matrix from the inertial to planet fixed frame, give an Earth reference. Earth sensors, magnetometers, and horizon sensors give an Earth-fixed reference, but are much less accurate than star and Sun sensors.

Gyros measure inertial rotation rates. Rate-integrating gyros give the integral of the body rates. Since each gyro integrates about a body axis, gyros cannot give an inertial reference directly and are always used in conjunction with magnetometers, star, Sun, or Earth sensors.

Angle encoders and potentiometers are used to measure relative orientations of components on the spacecraft. Encoders have potentially very high resolution but are expensive. Potentiometers, resistors whose output varies with angle, are generally cheaper than encoders and are often used for “sanity checks” on components driven by stepping motors. The stepping-motor position is usually known from counting steps but it is possible to lose count, for example at spacecraft separation from the launch vehicle.

Selection of sensors is a process of trading cost against sensor life, accuracy, mass, power consumption, and processing-software complexity. An important factor in selecting sensors is their radiation hardness. Radiation will degrade many analog circuits

and most CCD, CMOS, or CID imaging sensors, thus limiting the life of optical sensors. Other considerations include sensor placement and interference with the payload.

4.5.8 Selecting processors

The four major requirements for processors are

1. Processing throughput;
2. Memory;
3. Radiation hardness;
4. Heritage.

The processor needs to have sufficient throughput to perform all the computations required by the spacecraft. It needs to have sufficient memory for all required software. For example, the original processors on the Space Shuttle did not have enough memory for all of a missions' software, so it was necessary to load in new software during the mission. Radiation hardness is a requirement for all but very low-Earth orbiting satellites. Criteria are both the total-dose rating and immunity to single-event upsets (SEU) and latch-up immunity. The latter is when the processor freezes and has to be rebooted.

The last criteria, heritage, is important since it is much less expensive to use a processor that you have already used before. Changing processors often means changing software-development tools, operating systems, and even languages. These kind of changes have a major impact on the cost and schedule of a satellite program.

4.5.9 Selecting delta-V engines

The selection of the delta-V engine has an impact on the mission time line, the spacecraft cost, and all of the subsystems on the spacecraft. By delta-V engine we usually mean engines that are specifically designed for orbit-change operations and not for attitude control. The three major types are

1. Solid motor;
2. Liquid motor;
3. Electric motor.

The first gets the spacecraft into orbit quickest but usually results in the largest orbital errors that must be corrected with other thrusters. A solid motor is usually large and requires substantial structure. The liquid motor usually requires several burns for orbit raising. This means that transfer-orbit operations take longer but the final orbit is more accurate. Also, liquid motors are more efficient than solid motors thus lowering the overall mass of the vehicle and reducing launch costs. Electric motors are the most efficient but require high power for high thrust and still may take months to put the spacecraft in its final orbit.

4.5.10 Selecting the station-keeping engines

If station keeping is required station-keeping rocket motors are needed. If large amounts must be done (e.g., for inclination control of a geosynchronous satellite requiring about 500 m/s for 10 years) then an electric motor may make sense. For small amounts, conventional bipropellant or monopropellant thrusters would be sufficient. For very precise station keeping small electric thrusters can be used.

4.5.11 Selecting the interfaces

If you have a flight computer you need to hook it into your sensors and actuators. A spacecraft might have hundreds of sensors and actuators when you include the thermal-control subsystem. The data rates for each device vary. A heater requires a single bit to turn it on or off but a camera might be sending a megabit each second for processing. Typical serial interfaces are

- Analog lines multiplexed to a common A/D converter;
- I2C a popular serial bus;
- Databus (e.g., MIL-STD-1553, CAN, SpaceWire, Ethernet);
- RS-232 – a 1960s standard for the serial communication of data;
- RS-422 – a replacement for RS-232 permitting higher speeds and with better noise immunity.

Most spacecraft have a combination of interfaces to accommodate specific hardware. For example, many sensors and actuators only provide RS-422 interfaces, so an all-SpaceWire architecture may not be feasible. It is common to have an entire circuit board dedicated to analog input and output lines. Many popular processors, such as Arduino, have numerous analog pins.

Parallel interfaces are also possible, such as direct memory access (DMA) or IEEE 1284. For example a camera may benefit from a parallel interface to the flight processor. DMA allows a device to access the computer memory directly, thus saving on processor overhead.

4.5.12 Cost

Most manufacturers wish to design a satellite that meets all requirements and costs as little as possible (unless they have a cost-plus, fixed-fee contract). The total cost of a spacecraft is composed of recurring and nonrecurring costs. Recurring costs are the costs associated with building each spacecraft. Nonrecurring costs are costs associated with designing a spacecraft or customizing an existing design. Each type of cost is composed of the cost of personnel and the cost of purchased products, facilities, and other nonpersonnel costs. Recurring costs are discussed in Table 4.2. Nonrecurring costs are described in Table 4.3.

Table 4.2 Recurring costs.

Cost	Description
Direct Cost of Components	Costs of components generally rise with the required lifetime of the component and the device's performance. The cost of integrating a component into a design must be added to this cost. Sometimes, a more expensive component is much less expensive to integrate.
Direct Cost of the Consumables	The cost of consumables (such as fuel and cryogenics) must include the cost of loading the consumable and the cost of storing the consumable onboard the spacecraft while it is on the ground.
Cost to Lift the System into Orbit	Mass is almost always at a premium on any spacecraft. The transportation cost must always be considered. The greater the required delta-V for the mission the higher the cost. This is more complicated than it seems at first since there are thresholds of mass where you must go to the next size up launch vehicle (like from a Delta to an Ariane) that then has a different price schedule. This cost can be lowered by sharing a launch vehicle or going up on a test launch. The latter was used by PanAmSat to get a free launch and get started in the commercial communications-satellite business.
Cost of Insurance	New designs cost more to insure than proven designs. The cost of insurance also varies with the launch vehicle. Insurance is quite expensive and is provided by specialized insurance companies.
Cost of Maintenance	Technical support for a customer can be very expensive, especially if it requires onsite support. In addition to the direct cost of the engineering support, the cost of losing the services of the engineers as they move to other assignments, or leave the company, must be considered. Training of new engineers to support spacecraft is rarely done in a methodical way and can lead to support problems.

Table 4.3 Nonrecurring costs.

Cost	Description
Cost to Design the Algorithms	This includes the design costs and the cost of all tools needed to generate the design.
Cost to Implement in Software	Includes software requirements generation, software design, coding, and test.
Cost to Document the System	Includes memos, reports, and user guides.